

authentication protocols of layer 2. It would then find out the location and availability of the required resource via the collective layer of APIs that is layer 3. Next, it would ask for an exchange of data between networks or computers (remember that a grid is a collective) via the Resource protocols of layer 2. Finally, your program would monitor the progress of initiated transaction, pointing out success or failure as appropriate, via the Resource protocols.

tein folding. Proteins are the building blocks of life; they define us via chromosomes and defend us via antibodies, performing every function in between. Diseases such as Alzheimer's and cancer are linked to proteins and their property of folding: a means to align selves into a functional unit. To understand this mysterious process, simulations are run on computers. Protein folding used to take 20 months to simulate. Thanks now to grid-crunching numbers over supercomputers thrown across borders, it can be done in a day, at 1 trillion operations per second. The Human Genome Project, an undeniable task of gargantuan data shuffling, required 500 million trillion sequence comparisons. Such a feat would not have been possible without the advanced computational infrastructure that a grid delivers.

CERN's European Laboratory for Particle Physics in Switzerland expects to produce several petabytes of data each year (think 20 million, four-drawer filing cabinets full of text and you are thinking in petabytes). Running through this mass of digits would require computing power greater than 20 teraFLOPS—IBM's ASCI White hardly touches the 12 teraFLOPS mark. Clearly, joint computational power is called for and CERN is looking for a grid solution to its dilemma.

For the scientific community, a grid allows access to greater computing power and faster networks, which presents an environment for better sharing of data across borders, thus allowing for better co-operation. A grid-enabled earthquake simulator hopes to divide the task between nations across continents: while one laboratory will use its supercomputer to analyse stress caused on the foundation of a quake-hit bridge, another located miles away would simultaneously concentrate on the effects that the

rest of the structure show; effectively reducing the total time for a complete analysis by a huge factor.

Cheaper than the whole

When terrorist strikes and clandestine security hacks factor in a need for greater data access and redundancy, a distributed network becomes an ideal choice. A national government can share vital data on terror groups between states and its policing authorities. Moreover, thanks to a non-central, peer-to-peer network, even if a section of such a grid is made useless, other parts can chip in and balance the load to assure continuity of service. Such a data exchange need not be limited to intelligence; it could also be used effectively to impart greater transparency to bureaucracy and governance. The dream of that elusive global village has to materialise with the advent of computing resources to our physical villages, bridging the divide with low-cost distributed networks. Finally, for cash-strapped governments that cannot afford singular supercomputers to simulate ecological or military experiments, distributed computing offers equivalent processing power at a much lesser cost.

Indeed, cost benefits have attracted multinational corporations to grid. Although less than 1 per cent of companies are using grid computing today, the number is expected to grow to 10 per cent by 2005 when large companies with more than 10,000 employees will take a 40 per cent chunk of the grid pie. The corporate community needs a homogeneous, business-to-business, data-rich environment to conduct commerce. It has to leverage existing resources and delay the need to purchase new infrastructure. A company with 600 grid-enabled desktop PCs can utilise all of them together as one computational platform, providing it with enough juice to tackle the world's 49th largest supercomputer.

Take the case of GlobeXplorer, a company that delivers imagery from satellites and aerial photography over the Internet and other media. It has used distributed computing to great effect: to reduce its image processing time from 1 terabyte a month to 1 terabyte a week by scavenging spare compute cycles from its workstations and servers.

Info The world's fastest supercomputer, the IBM ASCI White, is rated at 12 TeraFLOPS and costs about \$110 million. Thanks to grid computing, SETI@home gets about 15 TeraFLOPS on an average and costs about \$500,000 to date

Greater than its parts

At this point in time the concept of grid computing is emerging from the labs of universities and government institutions, looking for a commercial helping hand that would allow it to reach a mass audience in a manner that the Internet did.

The scientific community has a voracious appetite for computational resources. Too much is never enough for those who trade data in terabytes on a daily basis. The work involved calls for no other solution but a distributed grid.

Take for instance, trying to understand the complexities involved in pro-